Topology: MAT307/M405/MAT803 : - Problem Set 2

1. Give an example of a T_1 space where a sequence converges to more than one point. Conclude that a T_1 space is not necessarily a Hausdorff Space.

2. Prove that the product topology $\mathcal{T}_{\text{prod}}$ on $\mathbb{R}^n$ (where each factor $\mathbb{R}$ is equipped with its standard Euclidean topology) coincides with the Euclidean topology $\mathcal{T}_{\text{Euc}}$ on $\mathbb{R}^n$ induced by the standard Euclidean metric.

3. Let $\{X_\alpha\}_{\alpha\in\Lambda}$ be a family of topological spaces, and let $\phi : \Lambda \longrightarrow \Lambda$ be a bijection. Prove that the product spaces $X = \prod_{\alpha\in\Lambda} X_\alpha$ and $X_\phi = \prod_{\alpha\in\Lambda} X_{\phi(\alpha)}$, both equipped with their respective product topologies, are homeomorphic.

4. Let $\{X_\alpha\}_{\alpha\in\Lambda}$ be a family of topological spaces, and for each $\alpha \in \Lambda$, let $Y_\alpha \subseteq X_\alpha$ be equipped with the subspace topology. Prove that the product topology on $Y = \prod_{\alpha\in\Lambda} Y_\alpha$ coincides with the subspace topology on Y induced by the product topology on $X = \prod_{\alpha\in\Lambda} X_\alpha$.

5. Let $\{X_\alpha\}_{\alpha\in\Lambda}$ be an indexed collection of topological spaces, and let $X = \prod_{\alpha\in\Lambda} X_\alpha$ be equipped with the product topology $\mathcal{T}_{\text{prod}}$. For any fixed index $\beta \in \Lambda$, consider the β-th coordinate projection map

$$p_\beta : \prod_{\alpha\in\Lambda} X_\alpha \longrightarrow X_\beta, \quad p_\beta((x_\alpha)_{\alpha\in\Lambda}) = x_\beta.$$

   Prove that p_β is an open map. Is p_β necessarily a closed map?

6. Let L be a line in the plane $\mathbb{R}^2$, and $\mathbb{R}_\ell$ be the lower limit topology on $\mathbb{R}$. Discuss the topology L inherits as a subspace of $\mathbb{R}_\ell \times \mathbb{R}$ and as a subspace of $\mathbb{R}_\ell \times \mathbb{R}_\ell$.

7. Compare (i) the ordered Square I_o^2, (ii) the topology I^2 inherits as a subspace of the product topology $\mathbb{R}^2$, and (iii) the product topology on $I \times I$.

8. Let X be a topological space, and $\Delta = \{(x, x) \in X \times X \mid x \in X\}$ be the diagonal of $X \times X$, equipped with the subspace topology inherited from the product topology on $X \times X$.
   (i) Prove that X is a Hausdorff if and only if the diagonal Δ is a closed subset of $X \times X$.
   (ii) Prove that the diagonal map $d : X \longrightarrow \Delta$ defined by $d(x) = (x, x)$ is a homeomorphism.

9. Let X and Y be topological spaces, and let $f, g : X \to Y$ be continuous maps. Assume that Y is a Hausdorff space. Prove the following statements:
   (i) The coincidence set $E = \{x \in X \mid f(x) = g(x)\}$ is closed in X.
   (ii) If D is a dense subset of X and $f|_D = g|_D$, then $f = g$ on all of X.

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(iii) The graph of f, defined as $\Gamma_f = \{(x, f(x)) \in X \times Y \mid x \in X\}$, is closed in the product space $X \times Y$.

(iv) Assume that $f : X \to Y$ is a continuous, open, and surjective map. Then the fiber relation set $R = \{(x_1, x_2) \in X \times X \mid f(x_1) = f(x_2)\}$ is closed in $X \times X$ if and only if Y is Hausdorff.

10. Let X be a topological space and $A \subset X$ be dense in X. Let Y be a Hausdorff space and $f : A \to Y$ be a continuous map. Show that if f may be extended to a continuous map $g : X \to Y$, then g is uniquely determined by f.

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